

Question 1

Luisa Kalkert, Luke Richard, Nicholas Chandler

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Introduction

In this section, we ask the question: “How does GDP per capita relate to selected quality of life and human-geographic metrics?”

The Human-Geographic information we have is:

[Info here]

The Quality of Life indicators are:

[Info here]

We begin the analysis by reading in the data (and filtering the relevant columns) as is done in the following R code.

We now concern ourselves with the cleaning of the selected columns as many have strings mixed in with numeric values.

Note that we have removed entries that may not be countries so our numbers may not reflect those of other surveys. The preprocessing is however, accessible.

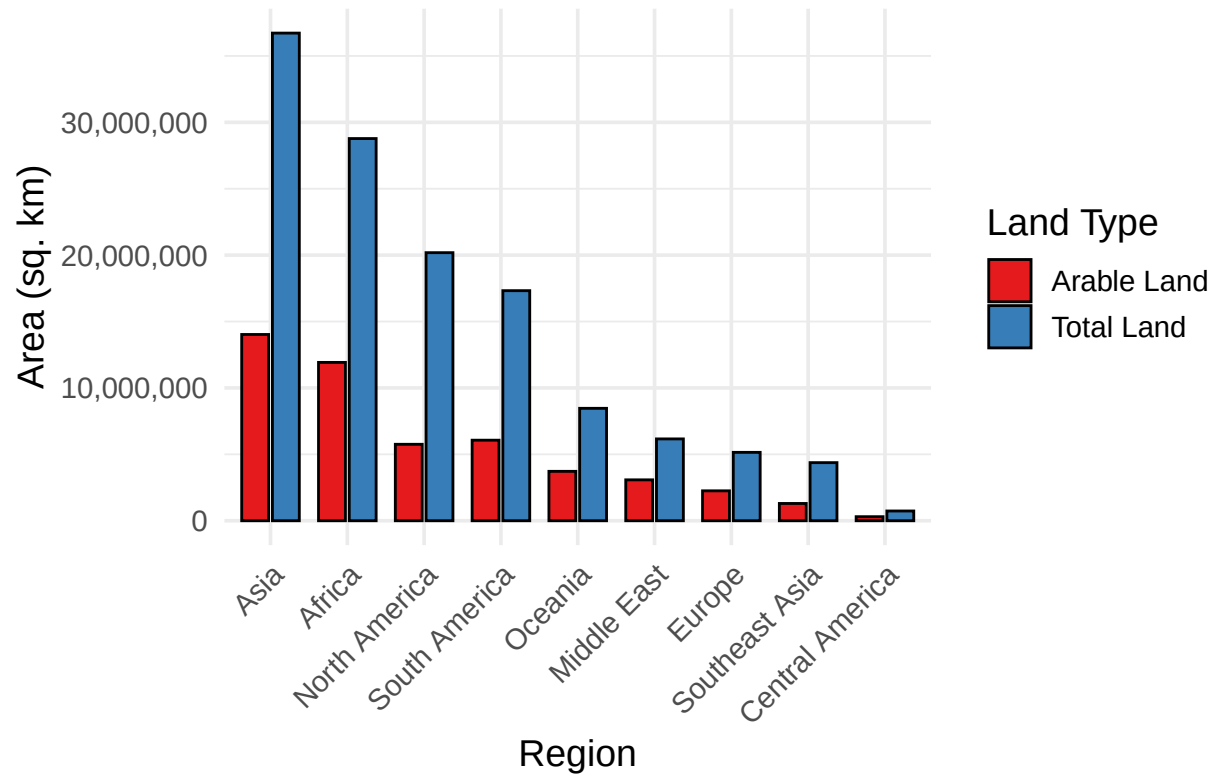
The continents we examine are:

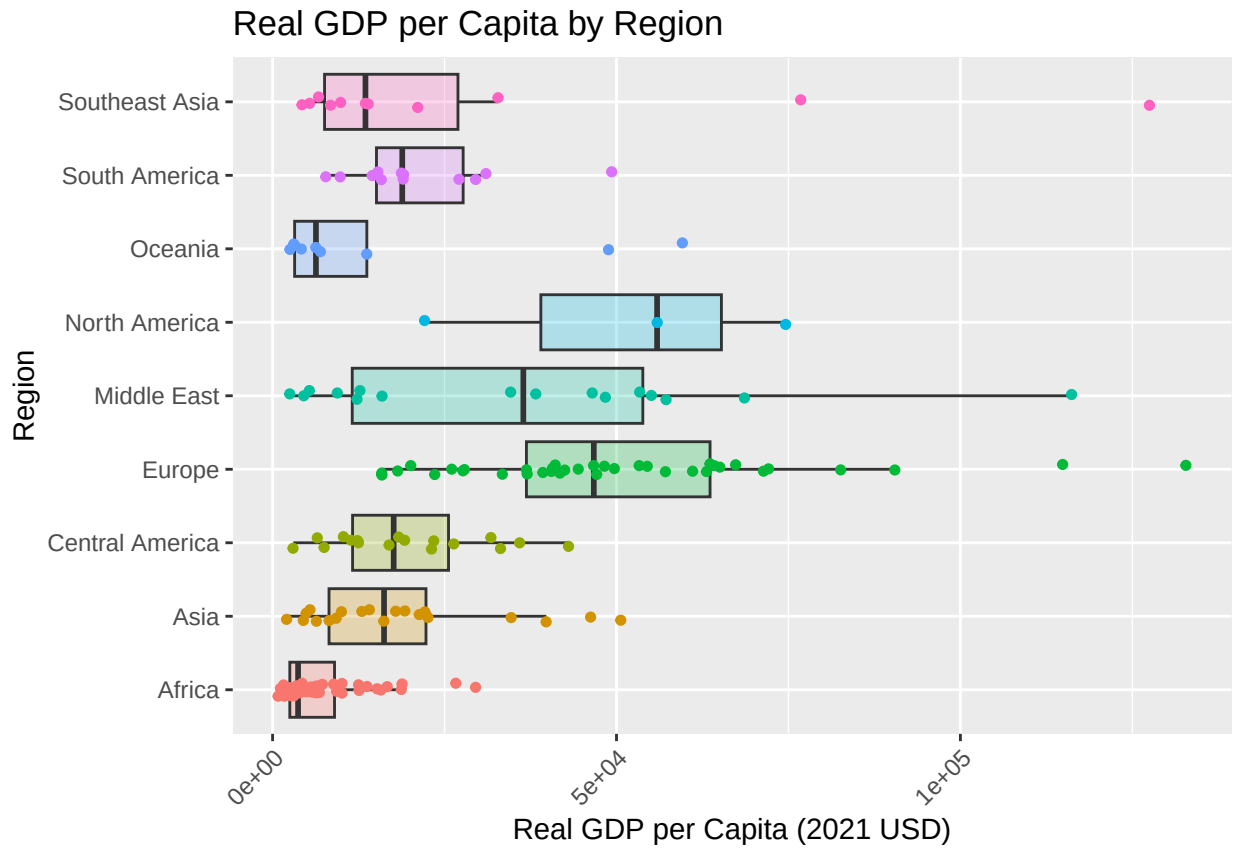
```
## [1] "Asia"           "Europe"          "Africa"           "Central America"
## [5] "South America"  "Oceania"         "Middle East"      "North America"
## [9] "Southeast Asia"
```

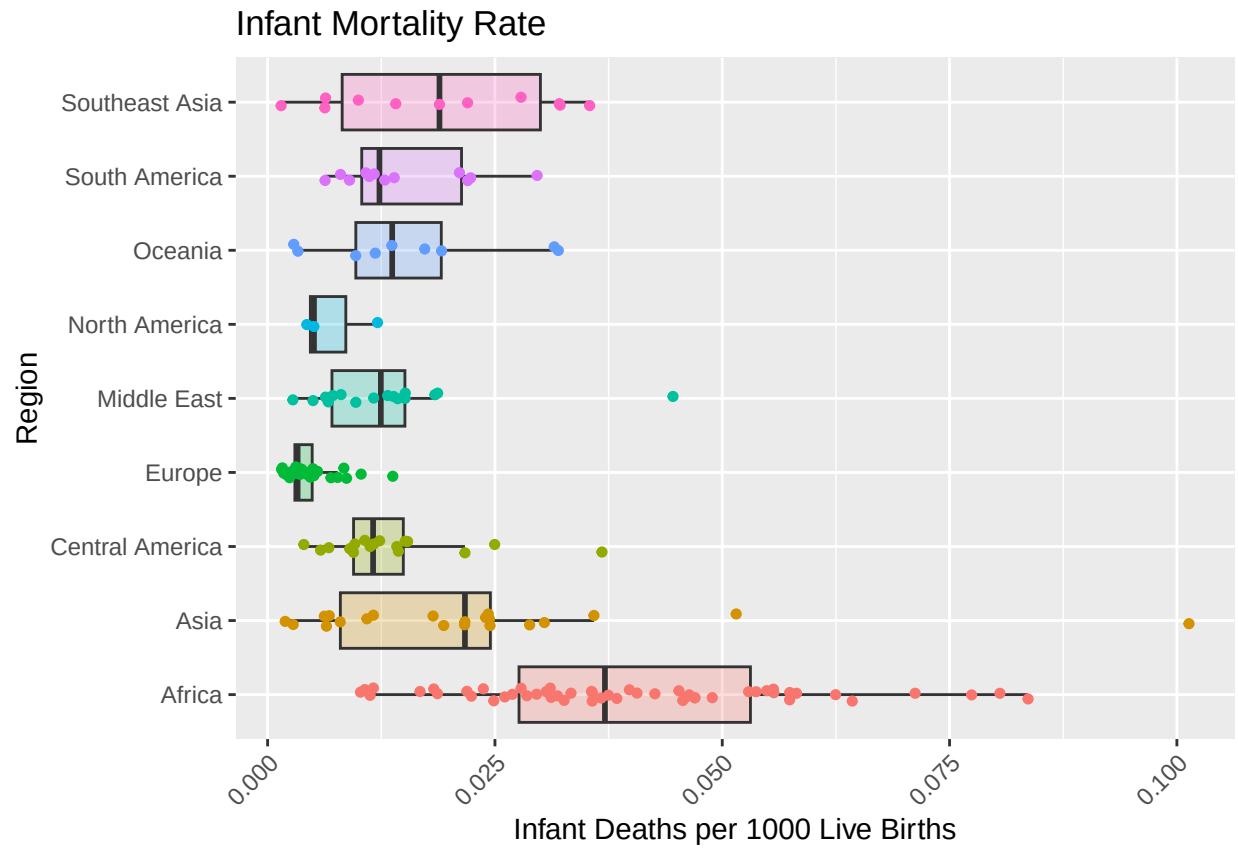
Exploratory Classical Graphics

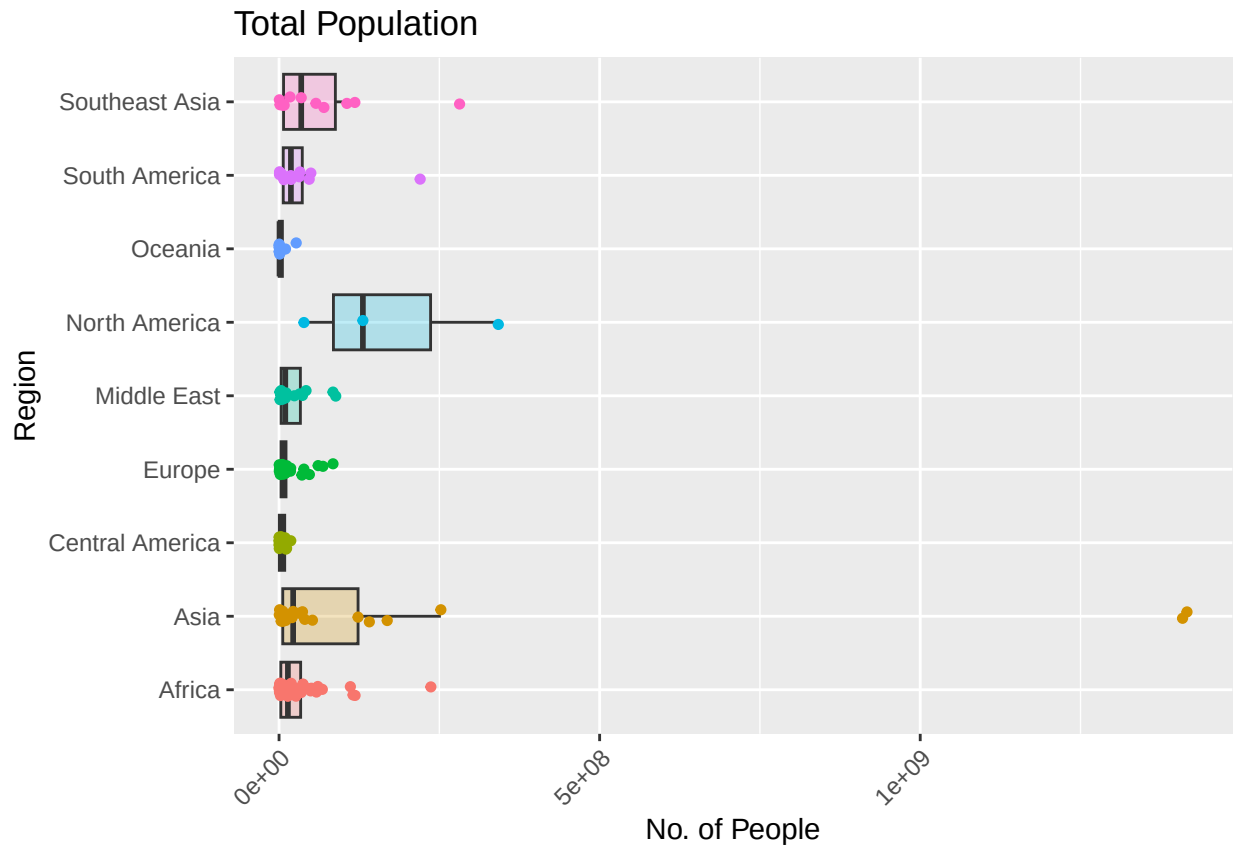
In this section, we examine a set of statistical graphics on a representative subset of the variables considered in the rest of this question. Specifically, we look at Land vs. Arable Land by region, GDP per capita, Infant Mortality Rate, and Total Population.

Land vs. Arable Land Area by Region









Correlation

In this section, we examine the correlation between selected continuous variables. Specifically, one can see that land, agricultural land, death rate, and total population have little correlation with other variables whereas the remaining variables seem to have complex correlation patterns among themselves.

Principal Components Analysis

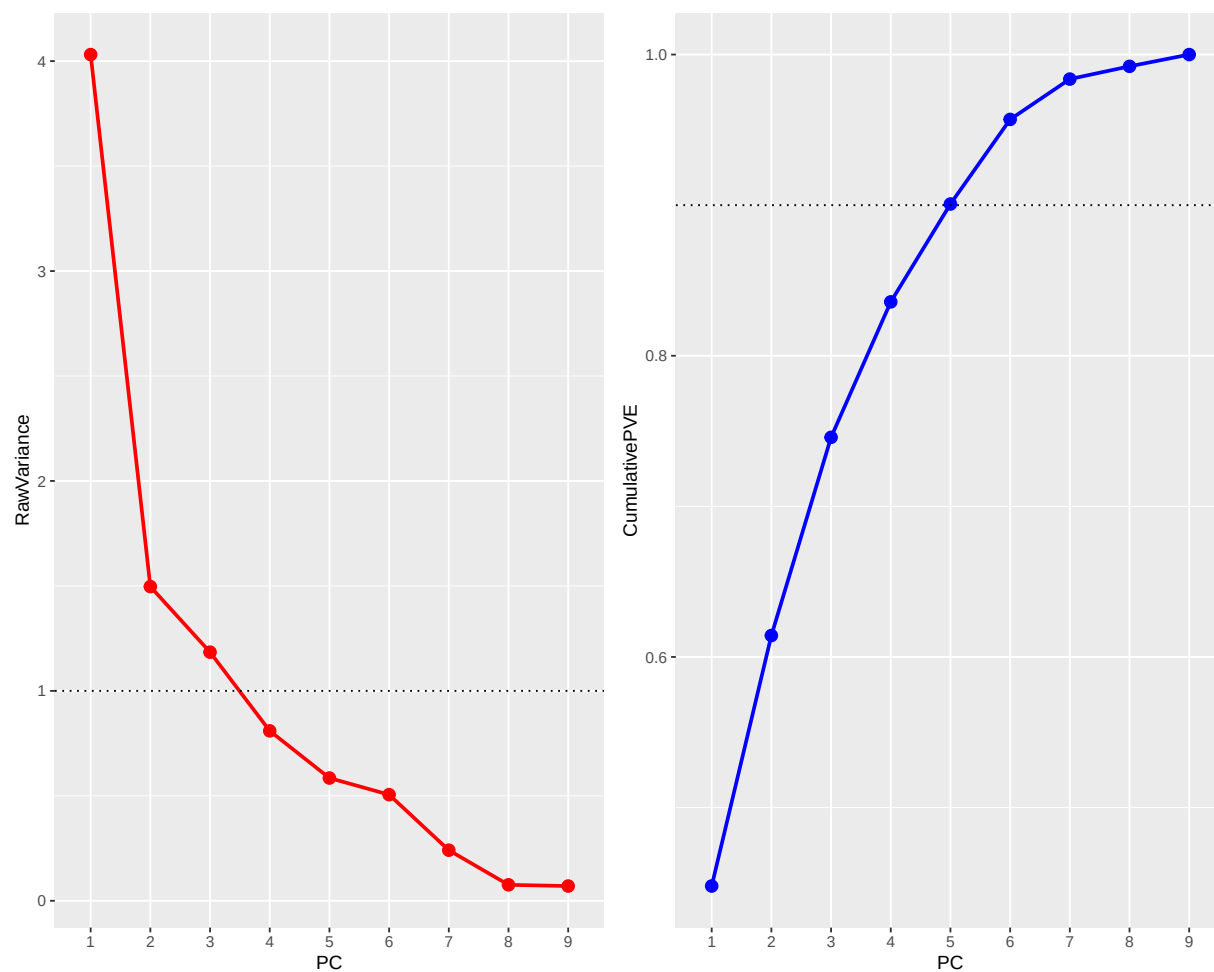
In this section we examine principal components analysis on the following variables:

```
## [1] "Total.sqkm"
## [3] "Real.GDP.per.capita.USD"
## [5] "Life.expectancy.at.birth"
## [7] "Live.births.per.1000.people"
## [9] "Total.population"

"Arable.land.percent"
"Infant.deaths.per.1000.live.births"
"Deaths.per.1000.people"
"Population.growth.rate.percent"

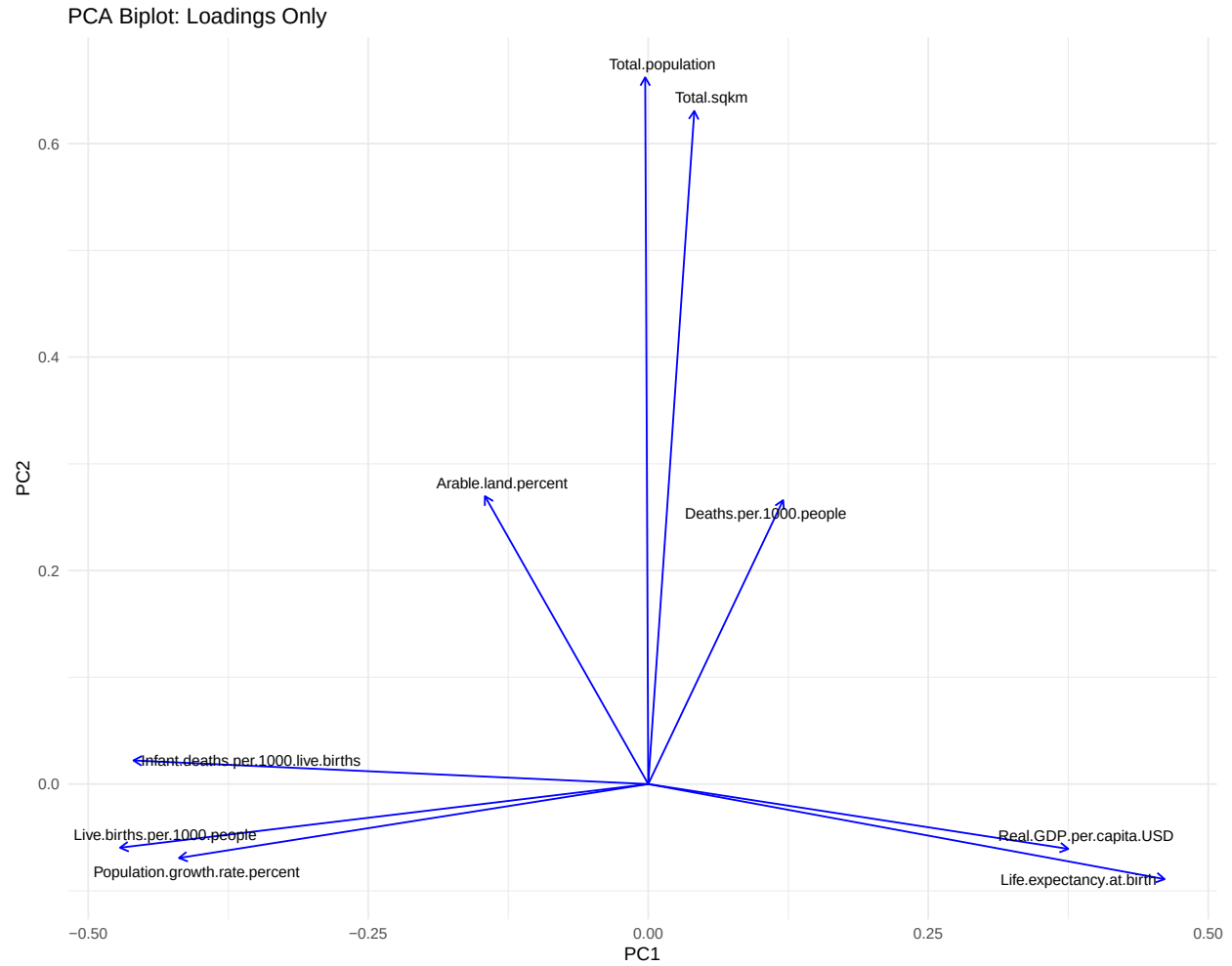
## NULL

## NULL
```



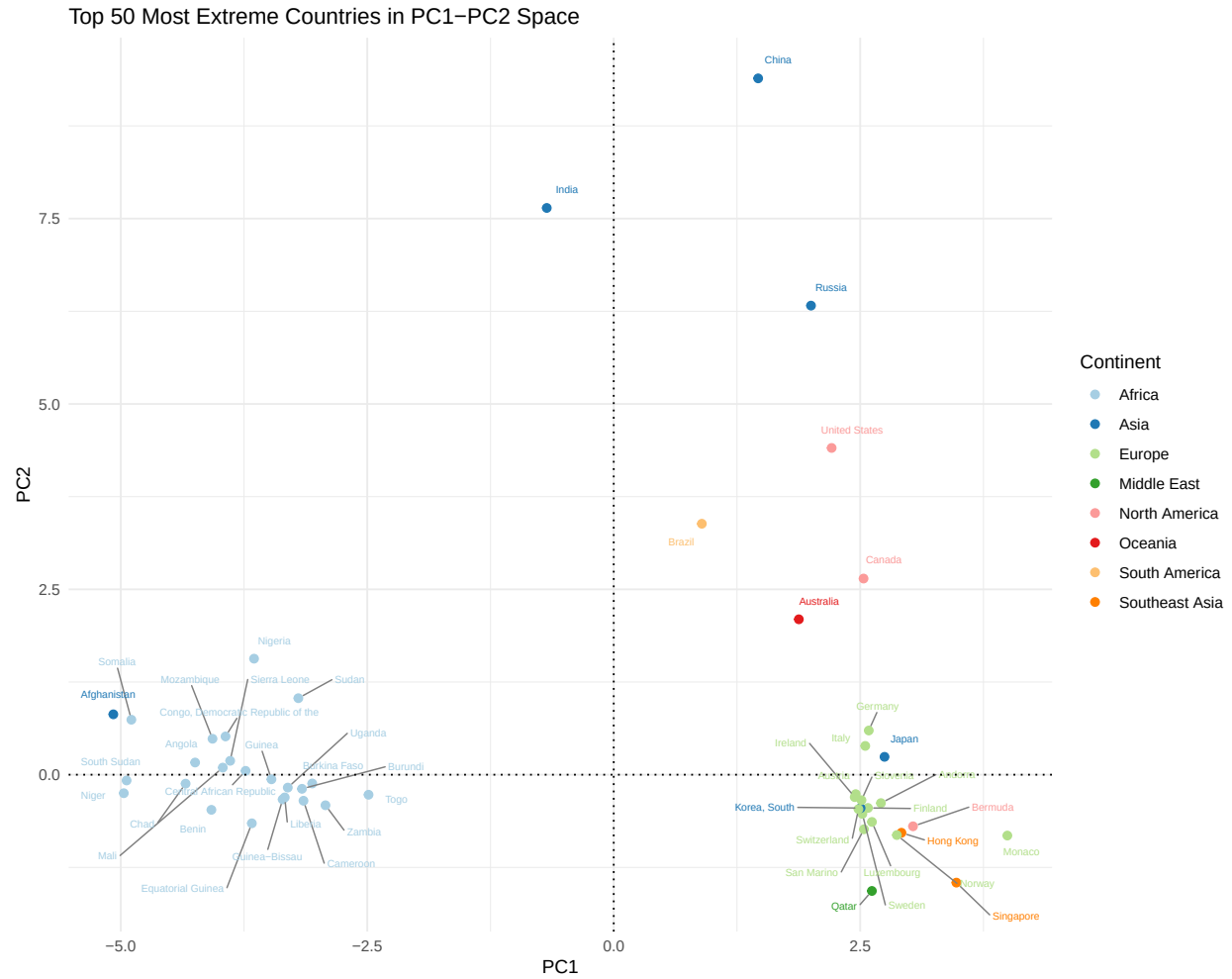
There are 3 principal components that “aggregate information” due to the variance being larger than 1 while the PCA was run on scaled data. Additionally, to capture ~90% of the variance, we need 5 PCs.

Here we plot the loading vectors of the PCA on the axes of the first and second principal components.



One can see that for small values of PC1 we see large values of Infant Mortality Rate, Birth Rate, and Population Growth Rate. For large values of PC1 we see large values of Real GDP per capita USD and Life Expectancy at Birth. One could interpret PC1 as the developedness of countries with larger values corresponding to more developed countries.

For large values of PC2 we see that the countries typically have large values of Total Population and Total SqKm and also (to a lesser degree) Arable Land and Death Rate. PC2 could be interpreted as the size of a country in terms of population and area.



We see groupings along the PC1 axis of more developed countries on the side with larger PC1 values and less developed countries with smaller PC1 values. There are several outliers with large values for PC2 that correspond with larger countries in terms of both population and land mass.